Two formulas on one line

The two partition coefficients of a species are the ratios of its concentrations in the phases,

$$P_A = \frac{c_A^{(1)}}{c_A^{(2)}} \qquad \text{and} \qquad P_B = \frac{c_B^{(1)}}{c_B^{(2)}}\,, \tag{1}$$

which can be measured. In a trial, suppose that

$$P(A = 1 \mid L = 1) = 0.75 \quad \text{and} \quad P(A = 1 \mid L = 0) = 0.50.$$

The weights must satisfy

$$0 < \pi(L_i) < 1 \quad \text{for all } i.$$

A long aligned row

Shifting the contour gives

$$\begin{aligned}\mathbb{E}[L_n] &= -\frac{1}{2\pi\mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} \frac{\widetilde{f}_n(s)}{s}\,\mathrm{d}s \\ &= -\frac{1}{2\pi\mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} \frac{1}{s^2(\psi(1-s)-\psi(1))}\,\mathrm{d}s + \frac{1}{2\pi\mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} \frac{\Gamma(s)\Gamma(n)/\Gamma(n+s)}{s(\psi(1-s)-\psi(1))}\,\mathrm{d}s.\end{aligned}$$

Lists that go on

The gaps are computed for $j = 1, 2, 3, \ldots$ obtained by diagonalisation, and the roots are ordered as $0 > s_1 > s_2 > \ldots$ denoting the negative roots.

Words that do not join

We have $x = 1$ and $y = 2$ in this case, and nothing else. When

$$a = b \qquad \text{where we assume} \qquad c = d,$$

the claim holds. Two numbered equations stay apart,

$$u = \frac{1}{2} \tag{2}$$

$$v = \frac{1}{3} \tag{3}$$

and so do two unrelated lines.

$p_1 = 3$ and $q_1 = 4$
is the tail of a sentence from the page before: it starts at the left margin and is no display. The rest of this page is running text, so that the column's margins can be measured against it, and it runs on for a few lines across the whole width of the column in the usual way.